

Risk prediction of major adverse cardiac events by high sensitivity troponin is modified by comorbidities

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PURPOSE

BACKGROUND

- High-sensitivity cardiac troponin (hs-cTn) is a biomarker for patients with possible acute coronary syndrome (ACS). Studies addressing the risk stratification between comorbidities and hs-cTn are limited.

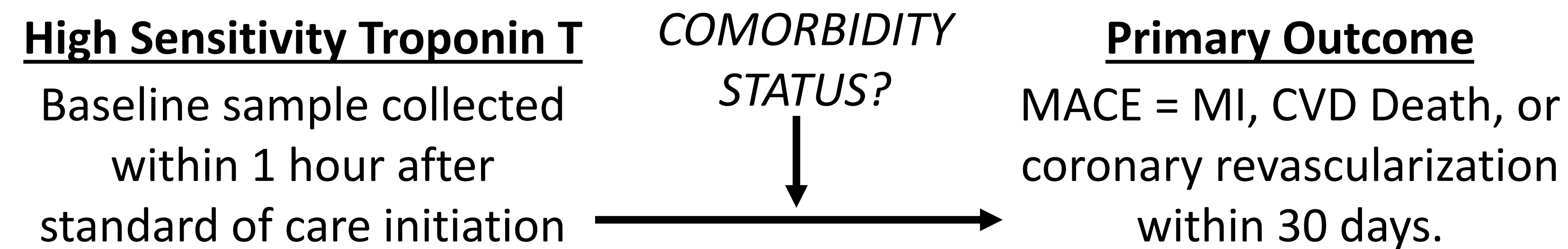
OBJECTIVE

- Determine whether comorbidities modify the relationship between baseline hsTnT values and the risk of 30-day major adverse cardiac events (MACE) in a multicenter United States (US) cohort.

METHODS

STUDY DESIGN

- Adult ED patients with suspected ACS were prospectively enrolled in a multicenter, US cohort study.
- Baseline hs-cTnT and patient characteristics were collected at enrollment. Primary outcomes at 30 days were adjudicated.



STATISTICAL ANALYSIS

- Hs-cTnT results were dichotomized by limit of quantification (LOQ) [6 ng/L] and upper reference limit (URL)[19 ng/L] (**Figure 1**)
- Association between LOQ and URL cut-offs with MACE was evaluated using logistic regression. Effect modification of comorbidity was assessed via statistical interaction (**Figure 2-3**)

COHORT CHARACTERISTICS

- Among 1460 participants included, 46.3% (676/1460) were female and 37.1% (542/1460) were Black with a mean age 57.6 ± 12.9 yrs.

RESULTS

Table 1: Summary of comorbidities in study population by outcome

Comorbidity, n (%)	With MACE (n=210, 14.4%)	Without MACE (n=1250, 85.6%)	Total (n=1460)
Hypertension (HTN)	161 (76.7)	809 (64.6)	970 (66.3)
Hyperlipidemia (HLD)	125 (59.5)	567 (45.3)	692 (47.3)
Diabetes Mellitus	84 (40.0)	347 (27.7)	431 (29.5)
Prior Coronary Artery Disease (CAD)	114 (54.3)	359 (28.7)	473 (32.4)
Prior Myocardial Infarction (MI)	82 (39.0)	236 (18.8)	318 (21.8)
Prior Percutaneous Coronary Intervention	65 (31.0)	181 (14.5)	246 (16.8)
Prior Coronary Artery Bypass Graft	39 (18.6)	66 (5.3)	105 (7.2)
Prior Cerebral Vascular Accident (CVA)	25 (11.9)	133 (10.6)	158 (10.8)
Prior Congestive Heart Failure (CHF)	68 (32.4)	238 (19.0)	306 (20.9)

Figure 1:

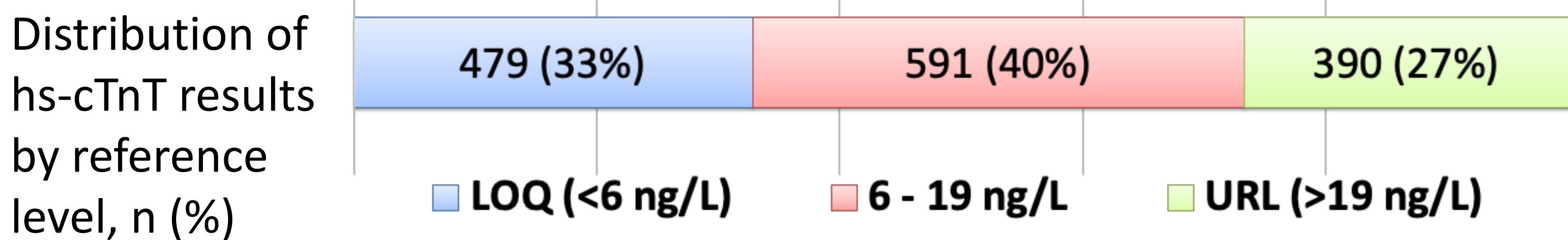


Figure 2: Forest plot of associations between baseline hs-cTnT < 6 ng/L [LOQ] with 30-Day MACE by selected comorbidities (P for interaction < 0.50)

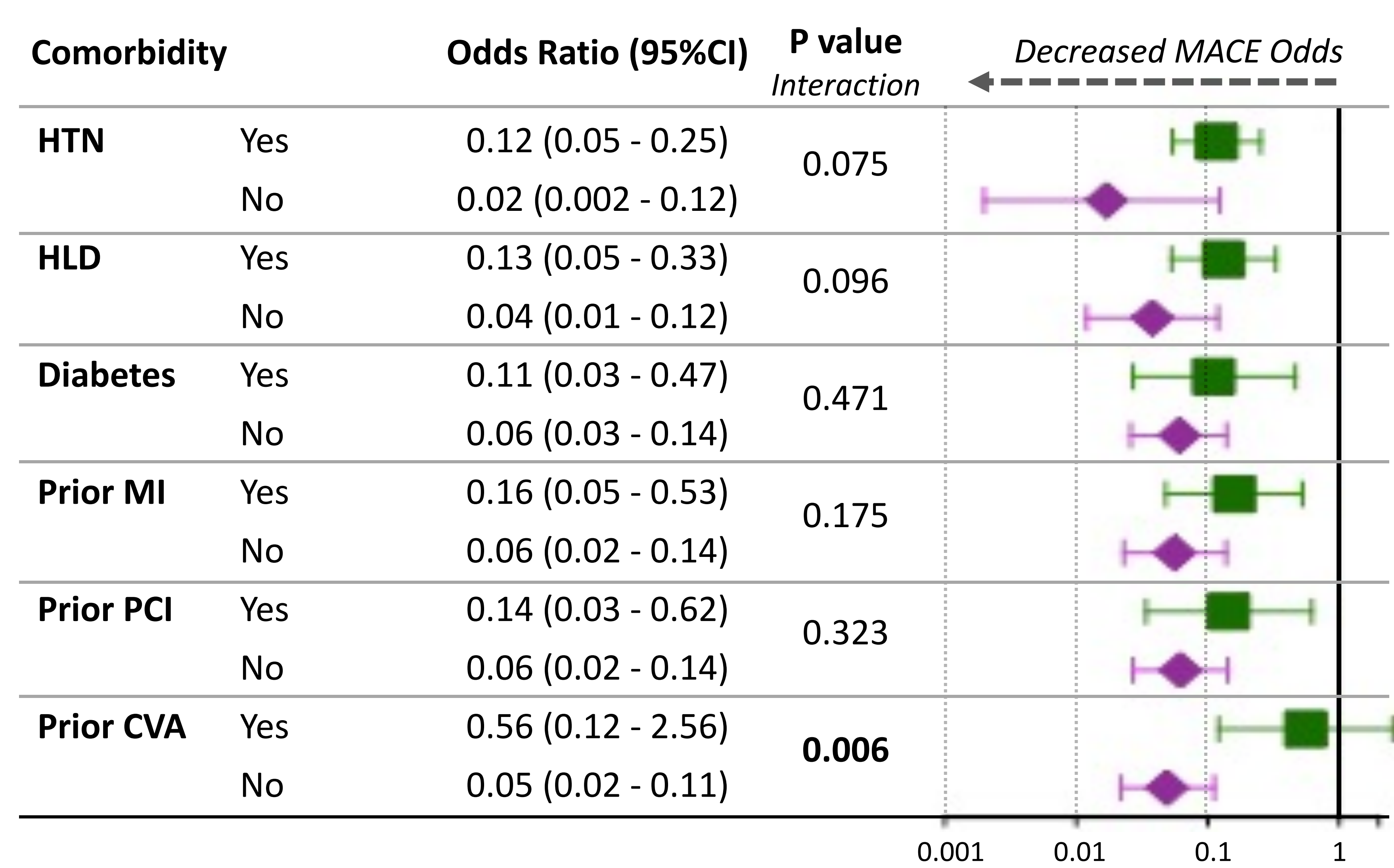
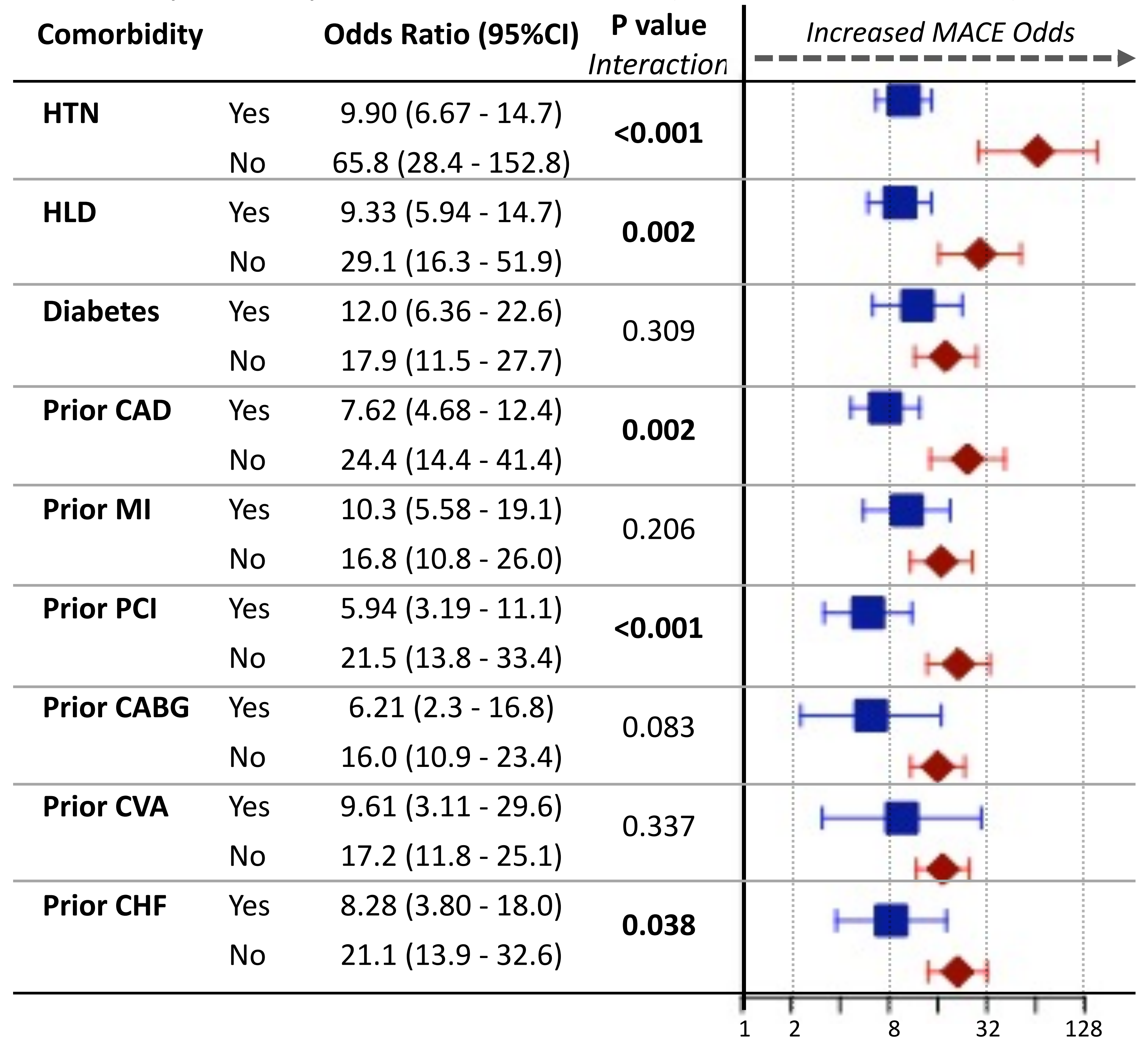


Figure 3: Forest plot of associations between baseline hs-cTnT ≥ 19 ng/L [URL] with 30-Day MACE by selected comorbidities (P for interaction < 0.50)



CONCLUSION

- In a diverse, multicenter, US cohort the association between hs-cTnT and MACE was **weakened** by the presence of comorbidity.
- Safety of hs-cTnT strategies may be diminished when applied to populations with a high comorbid prevalence
- Future research will assess efficacy of diagnostic risk classification after stratification by comorbid status